

EC107: Microeconomics

In-class Exercises 4 — Government Policy & the Efficiency of Markets

Covers Days 5–6. For each **concept check**: vote, discuss with a neighbour, vote again. Short exercises are timed (6 minutes each).

A. Concept checks

- C1.** The equilibrium rent for a flat is £1,000/month. The government sets a rent *ceiling* of £1,200. The result is:
 (a) a shortage (b) a surplus
 (c) no effect (non-binding) (d) an immediate black market
- C2.** A £2 per-unit tax is, by law, placed on *sellers* rather than buyers. Compared with placing it on buyers, the quantity traded and the prices buyers pay / sellers keep will be:
 (a) better for sellers (b) better for buyers
 (c) exactly the same (d) dependent on who is wealthier
- C3.** The deadweight loss of a tax is best described as:
 (a) the revenue the government collects
 (b) surplus destroyed that reaches no one
 (c) always larger than the revenue raised
 (d) zero whenever the tax is small
- C4.** A market allocation is *efficient* when:
 (a) the outcome is fair (b) total surplus is maximised
 (c) everyone receives an equal share (d) the regulator approves
- C5.** For a tax of a given size, the deadweight loss is *largest* when:
 (a) both demand and supply are very inelastic
 (b) both demand and supply are very elastic
 (c) the tax is levied on sellers (d) revenue happens to be zero

B. Short exercises

Use the market Demand: $P = 30 - Q$, Supply: $P = 6 + Q$ for both questions.

- N1. (A tax: incidence, revenue, deadweight loss)** A per-unit tax of £8 is imposed.
- (i) Find the pre-tax equilibrium price and quantity.
 - (ii) With the tax, find the quantity traded, the price buyers pay P_b , and the price sellers keep P_s .
 - (iii) Compute the government's revenue.
 - (iv) Compute the deadweight loss, and state how the £8 burden splits between buyers and sellers.
- N2. (Welfare & a price ceiling)**
- (i) At the (pre-tax) equilibrium, compute consumer surplus, producer surplus and total surplus.
 - (ii) A price *ceiling* of £12 is now imposed. How many units are traded?
 - (iii) Compute the deadweight loss caused by the ceiling.

C. Discussion

- D1.** After a hurricane, the price of bottled water triples. A politician proposes an anti-price-gouging law capping the price. Give the strongest *efficiency* argument against the cap and the strongest *equity* argument for it. What can economic analysis tell us here, and what can it *not* tell us?